



CL 299 presentation 7

LNP Manufacturing and Process Simulation

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Energy

Syringe Pump :

Flow Rate	Power
1–10 mL/min	~10 W
10–50 mL/min	~15–20 W

Peristaltic Pump :

Flow Rate	Power
10–50 mL/min	20–40 W
50–100 mL/min	40–80 W

TFF circulation pump :- 80–120 W

OverHead Stirrer :- 50–100 W

Magnetic stirrer :- 5–15 W

Total power :- 695W

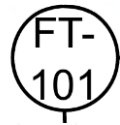
PAT AS P&ID

Some elements that we use in the P&ID.

- Level Indicator(LI)



- Flow Transmitter



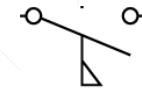
- Valve



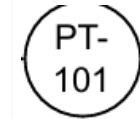
- Temperature transmitter



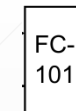
- Low flow valve



- Pressure transmitter



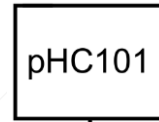
- Flow controller



- Conductivity sensor



- pH Controller



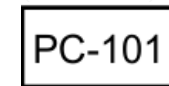
- pH transmitter



- Emergency Shutdown (ESD)



- Pressure controller



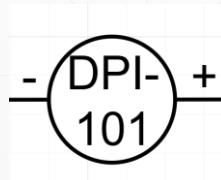
- Process line



- Control loop line



- Differential Pressure Indicator



Why there is no Cooling for Mixer?

- We have Low flow rate
- There is not much increment in temperature at lab scale

Resource

- P. R. Cullis and M. J. Hope, “Lipid nanoparticle systems for enabling gene therapies,” *Molecular Therapy*, 2017
- K. A. Whitehead *et al.*, “Formulations and methods for delivery of nucleic acids,” US Patent US20180208676A1, 2018
- J. R. Bourne and R. G. K. Friberg, “Mixing apparatus and method,” US6874900B2, Apr. 5, 2005.
- J. R. M. Clark *et al.*, “Bioreactor mixing system,” US20110223636A1, Sep. 15, 2011.
- D. J. McCarthy, “Agitation system with impeller,” US20100060325A1, Mar. 11, 2010.

THANK YOU!!!